

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: CRYPTOGRAPHY AND NETWORK SECURITY
COURSE CODE: CSA51

COURSE OUTCOME COVERED

CO2: Analyze Public Key crypto system strategies using RSA and key Exchange for Encryption algorithm to strengthen information security. (BL4,BL5)

Scenario-Based : 20%

QUESTIONS:

Total Marks: 50

Annexure A

Code of Conduct:

I Dilliraj N (192324249)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Dilliraj N
Signature of the student:

Annexure - A

Scenario-based

1. Secure Data Transmission using Block Cipher

A company is transmitting repeated financial data blocks over a network. Initially, they used ECB mode and noticed patterns in ciphertext.

A. Explain why this issue occurs.

Electronic Codebook (ECB) mode encrypts each plaintext block **independently** using the same encryption key.

- If two plaintext blocks are **identical**, ECB produces **identical ciphertext blocks**.
- As a result, repeated patterns in the plaintext remain visible in the ciphertext.
- An attacker can identify these patterns and gain information about the original data, even without knowing the encryption key.
- This makes ECB **unsuitable for transmitting sensitive or repetitive data**, such as financial records.

Example:

- Plaintext: Salary = \$5000 | Salary = \$5000
- ECB Ciphertext: X1 | X1 (same ciphertext block repeated)

B. Suggest a more secure block cipher mode and justify your choice based on the scenario.

Justification:

- CBC uses an **Initialization Vector (IV)** and XORs each plaintext block with the previous ciphertext block before encryption.
- Even if two plaintext blocks are identical, their ciphertext blocks will be **different** because each block depends on the previous encrypted block.
- This removes visible patterns from the ciphertext and provides much better confidentiality than ECB.

Advantages of CBC:

- Hides repeating patterns in plaintext.
- More secure for transmitting financial and confidential data.
- Widely used for secure block encryption (when implemented with a random IV and proper integrity protection).

2. Choosing an Encryption Algorithm for Banking System

A banking application needs high security and moderate performance for encrypting customer transactions. The available options are DES, 3-DES, and Blowfish.

- A. Identify the most suitable algorithm for this scenario.
- B. Justify your choice based on security strength and efficiency.

☐ **High Security:** Blowfish uses a **64-bit block size** and supports **variable key lengths from 32 to 448 bits**, making it much more secure than DES and generally stronger than 3-DES in practical use.

☐ **Better Performance:** Blowfish is **faster than 3-DES** because it performs fewer encryption operations while maintaining strong security.

☐ **DES Limitation:** DES uses a **56-bit key**, which is vulnerable to brute-force attacks and is no longer considered secure.

because it encrypts data three times, making it less efficient for banking transactions.

☐ **Suitable for Banking:** Blowfish provides a good balance of **strong security** and **moderate performance**, making it the best choice among the given options.

3. Key Exchange in an Unsecured Network

Two users want to establish a secure communication channel over a public network using Diffie-Hellman key exchange. However, the network is vulnerable to attacks.

- A. Explain how the key exchange works in this scenario.

☐ Both users publicly agree on a **prime number (p)** and a **generator (g)**.

☐ User A chooses a **private key (a)** and computes the public key: $A = g^a \bmod p$.

☐ User B chooses a **private key (b)** and computes the public key: $B = g^b \bmod p$.

☐ They exchange their public keys over the public network.

☐ User A computes the shared secret: $K = B^a \bmod p$.

☐ User B computes the shared secret: $K = A^b \bmod p$.

☐ Both obtain the **same shared secret key**, which is then used for secure encryption.

- B. Identify a possible security threat and suggest a solution to mitigate it.

Threat:

- **Man-in-the-Middle (MITM) Attack:** An attacker intercepts the exchanged public keys and establishes separate keys with both users, allowing the attacker to read or modify the communication.

Solution:

- **Authenticate the key exchange** using **Digital Signatures** or **Digital Certificates (PKI)** to verify the identity of both users.
- This prevents attackers from impersonating either party during the key exchange.

4. Public Key Cryptography for Email Security

An organization plans to implement RSA for secure email communication.

- A. Describe how RSA can be used for encryption and decryption in this scenario.

RSA for Encryption and Decryption

1. Each user generates an **RSA key pair**:
 - **Public Key** – Shared with others.
 - **Private Key** – Kept secret by the owner.
2. The sender encrypts the email using the **recipient's public key**.
3. The encrypted email is sent over the network.
4. The recipient decrypts the email using their **private key**.
5. Since only the recipient has the private key, only they can read the message.

- B. Explain how key distribution and management play a role in maintaining security.

Key Distribution and Management

- **Public keys** must be distributed securely so users can trust they belong to the correct person.
- **Digital Certificates** issued by a **Certificate Authority (CA)** are used to verify the authenticity of public keys.
- **Private keys** must be stored securely and never shared.

- Regular **key renewal, revocation, and backup** help maintain long-term security.

5. Cryptography for IoT Devices

A smart home system uses low-power IoT devices that need secure communication. The designer must choose between RSA and Elliptic Curve Cryptography (ECC).

- A. Analyze the constraints of IoT devices.

Constraints of IoT Devices

- **Limited processing power** – IoT devices have low-performance processors.
- **Limited memory and storage** – They cannot handle large cryptographic operations efficiently.
- **Low battery power** – Encryption should consume minimal energy.
- **Limited bandwidth** – Smaller keys and messages are preferred for faster communication.
- **Real-time communication** – Fast encryption and decryption are required.

- B. Recommend a suitable cryptographic approach and justify your answer.

Recommendation and Justification

Recommended Approach: Elliptic Curve Cryptography (ECC)

Justification:

- ECC provides **strong security with much smaller key sizes** than RSA.
- Smaller keys require **less processing power, memory, and battery**, making ECC ideal for resource-constrained IoT devices.
- ECC offers **faster key generation and key exchange** with lower communication overhead.
- RSA requires much larger keys (e.g., **2048 bits**) to achieve comparable security, making it less efficient for low-power IoT devices.

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand